

Saurabh Vishwakarma

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INTRODUCTION

AI/ML enthusiast with hands-on experience in building and deploying machine learning applications using FastAPI and Flask. Strong foundation in statistics, linear algebra, and model evaluation with a focus on solving real-world problems through data-driven approaches. Passionate about developing scalable systems and continuously improving model performance through analytical experimentation.

PROJECTS

ATS Intelligence Engine (NLP + Docker) 2026

Python, FastAPI, Sentence Transformers, PyMuPDF, Docker

Built an AI-powered web application to evaluate resumes against job descriptions using semantic similarity and keyword matching. Extracted and processed text from PDF files using PyMuPDF, implemented Sentence Transformers for contextual similarity scoring, and designed a keyword-based scoring mechanism for skill overlap. Generated ATS scores with actionable insights and deployed a scalable REST API using FastAPI with Docker-based containerization.

Language Modeling using GRU 2025

Python, PyTorch

Developed a GRU-based sequence model with a focus on analytical evaluation by studying token distributions and sequence patterns prior to training. Monitored model performance using loss curves and perplexity trends, and conducted epoch-wise error analysis to improve sequence prediction quality and training stability.

House Price Prediction Platform (Deployed) 2025

Python, Pandas, NumPy, Scikit-learn, Flask

Developed a data-driven regression platform by performing comprehensive exploratory data analysis to identify key pricing factors, feature distributions, and multicollinearity. Applied statistical validation techniques including correlation analysis and residual diagnostics, followed by feature engineering and model evaluation using RMSE and cross-validation. Deployed the predictive model as a Flask-based web application for real-time inference.

EDUCATION

Chandigarh University 2023 – 2026

BCA — CGPA: 8.03

IIT Madras 2023 – 2027

BS in Data Science — CGPA: 7.2

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Pandas, NumPy, Matplotlib, Seaborn

ML Frameworks: Scikit-learn, PyTorch, Sentence Transformers

Core ML: EDA, Feature Engineering, Regression, Classification, Model Evaluation, Error Analysis

Statistics: Probability, Hypothesis Testing, Correlation, Bias-Variance Tradeoff

Deployment: FastAPI, Flask, Docker, REST APIs, MLflow (basic), Model Monitoring

Concepts: Structured Problem Solving, System Thinking, Optimization

Math: Linear Algebra, Probability Theory

CERTIFICATIONS

Foundation in Data Science – IIT Madras

Advanced Python Programming – Coursera

Generative AI Fundamentals – Coursera